# Theory

Let me first introduce notations in Table 1.

Table 1

| Notation | Meaning |
|---|---|
| $i$ | Industry sector |
| $P_i$ | The price of the goods produced by sector $i$ |
| $Y_i$ | Output (value added) of sector $i$ |
| $Y = \sum_i P_i Y_i$ | Total value added (GDP) in the economy |
| $\chi_i = \frac{P_i Y_i}{Y} = \frac{P_i Y_i}{\sum_i P_i Y_i} = \frac{\text{GDP}_i}{\text{GDP}}$ | The share of sector $i$'s GDP |
| $W_i$ | Wage per worker in sector $i$ |
| $L_i$ | Number of workers in sector $i$ |
| $W_i L_i$ | Total wage bill in sector $i$ |
| $WL = \sum_i W_i L_i$ | Total wage bill in the economy |
| $\ell_i = \frac{W_i L_i}{WL}$ | The share of the wage bill in sector $i$ |
| $s_i^L = \frac{W_i L_i}{P_i Y_i} = \frac{\text{Total wage bill}_i}{\text{GDP}_i}$ | The labor share in sector $i$ |
| $s^L = \frac{WL}{Y} = \frac{\text{Total wage bill}}{\text{GDP}}$ | The labor share in the economy |
| $\Gamma_i = \Gamma(N_i, I_i)$ | The task content of production with regards to labor in sector $i$ |
| $\gamma_i^L$ | The comparative advantage schedules for labor in sector $i$ |
| $\gamma_i^K$ | The comparative advantage schedules for captial in sector $i$ |

The production equation that will be used throughout Acemoglu and Restrepo (2019) is below. This equation is provided in the online appendix (it does not appear in the main paper).

(A2)
$$Y = \left\{ \left( \int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\, dz \right)^{\frac{1}{\sigma}} (A^K K)^{\frac{\sigma-1}{\sigma}} + \left( \int_{I}^{N} \gamma^L(z)^{\sigma-1}\, dz \right)^{\frac{1}{\sigma}} (A^L L)^{\frac{\sigma-1}{\sigma}} \right\}^{\frac{\sigma}{\sigma-1}}$$

**The task content of production with regards to *labor*** is defined as below.

(A3)
$$\Gamma \equiv \frac{\int_I^N \gamma^L(z)^{\sigma-1}\, dz}{\int_{N-1}^I \gamma^K(z)^{\sigma-1}\, dz + \int_I^N \gamma^L(z)^{\sigma-1}\, dz}$$

Therefore, **the task content of production with regards to *capital*** is defined as $1 - \Gamma$. Meanwhile, **Total Factor Productivity (TFP)** is defined as below.

(A3-2)
$$\Pi \equiv \left\{ \int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\, dz + \int_{I}^{N} \gamma^L(z)^{\sigma-1}\, dz \right\}^{\frac{1}{\sigma-1}}$$

Rearranging Equation (A2) yields Equation (A2-2), which appears in the page 8 of the main paper.

$$
\begin{aligned}
Y =&\Bigg\{ \Big( \int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz + \int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz \Big)^{\frac{1}{\sigma}} \Big( \frac{\int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz}{\int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz + \int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz} \Big)^{\frac{1}{\sigma}} (A^K K)^{\frac{\sigma-1}{\sigma}} \\
&+ \Big( \int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz + \int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz \Big)^{\frac{1}{\sigma}} \Big( \frac{\int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz}{\int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz + \int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz} \Big)^{\frac{1}{\sigma}} (A^L L)^{\frac{\sigma-1}{\sigma}} \Bigg\}^{\frac{\sigma}{\sigma-1}} \\
=&\Big( \int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz + \int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz \Big)^{\frac{1}{\sigma-1}} \\
&\Bigg\{ \Big( \frac{\int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz}{\int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz + \int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz} \Big)^{\frac{1}{\sigma}} (A^K K)^{\frac{\sigma-1}{\sigma}} \\
&+ \Big( \frac{\int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz}{\int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz + \int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz} \Big)^{\frac{1}{\sigma}} (A^L L)^{\frac{\sigma-1}{\sigma}} \Bigg\}^{\frac{\sigma}{\sigma-1}}
\end{aligned}
$$

(A2-2)

$$
=\Pi \Bigg\{ \Big(1-\Gamma\Big)^{\frac{1}{\sigma}} (A^K K)^{\frac{\sigma-1}{\sigma}} + \Big(\Gamma\Big)^{\frac{1}{\sigma}} (A^L L)^{\frac{\sigma-1}{\sigma}} \Bigg\}^{\frac{\sigma}{\sigma-1}}
$$

Another definition that appears in the main paper is **the labor share**, which is the wage bill (WL) divided by value added (Y) as below:

(A4) $$s^L \equiv \frac{WL}{Y}, \text{ where } Y \text{ is Equation (A2)}$$

To acquire this labor share, we need to know $W$, but it is explained neither in the main paper nor in the online appendix. $W$ can be acquired by solving the profit maximization problem as below:

$$
\begin{aligned}
&\max_{L,K}(Y - WL - RK), \text{ where } Y \text{ is Equation (A2)}\\
&\Rightarrow W = \frac{\sigma}{\sigma-1}\Bigg\{\Big(\int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz\Big)^{\frac{1}{\sigma}}(A^K K)^{\frac{\sigma-1}{\sigma}}\\
&+\Big(\int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz\Big)^{\frac{1}{\sigma}}(A^L L)^{\frac{\sigma-1}{\sigma}}\Bigg\}^{\frac{1}{\sigma-1}}\Big(\int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz\Big)^{\frac{1}{\sigma}}\\
&\times (A^L)^{\frac{\sigma-1}{\sigma}}(\frac{\sigma-1}{\sigma})L^{\frac{-1}{\sigma}}
\end{aligned}
\tag{W}
$$

Also,

$$
\begin{aligned}
&\Rightarrow R = \frac{\sigma}{\sigma-1}\Bigg\{\Big(\int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz\Big)^{\frac{1}{\sigma}}(A^K K)^{\frac{\sigma-1}{\sigma}}\\
&+\Big(\int_{I}^{N} \gamma^L(z)^{\sigma-1}\,dz\Big)^{\frac{1}{\sigma}}(A^L L)^{\frac{\sigma-1}{\sigma}}\Bigg\}^{\frac{1}{\sigma-1}}\Big(\int_{N-1}^{I} \gamma^K(z)^{\sigma-1}\,dz\Big)^{\frac{1}{\sigma}}\\
&\times (A^K)^{\frac{\sigma-1}{\sigma}}(\frac{\sigma-1}{\sigma})K^{\frac{-1}{\sigma}}
\end{aligned}
\tag{R}
$$

Plugging this $W$ into Equation (A4) yields Equation (A5) as below, which appears in the online apendix:

$$
s^L = \frac{1}{1+\left(\frac{1-\Gamma}{\Gamma}\right)^{\frac{1}{\sigma}}\left(\frac{A^K K}{A^L L}\right)^{\frac{\sigma-1}{\sigma}}}
\tag{A5}
$$

However, the main paper does not use this Equation (A5). Instead, it uses Equation (A4). The following process leads to Equation (A4). From Equation (W) and (R),

$$
\begin{aligned}
\frac{W}{R} &= \left(\frac{\Gamma}{1-\Gamma}\right)^{\frac{1}{\sigma}}\left(\frac{A^L}{A^K}\right)^{\frac{\sigma-1}{\sigma}}\left(\frac{K}{L}\right)^{\frac{1}{\sigma}}\\
&\Rightarrow \left(\frac{K}{L}\right)^{\frac{\sigma-1}{\sigma}} = \left(\frac{W}{R}\right)^{\sigma-1}\left(\frac{1-\Gamma}{\Gamma}\right)^{\frac{\sigma-1}{\sigma}}\left(\frac{A^K}{A^L}\right)^{\frac{(\sigma-1)^2}{\sigma}}
\end{aligned}
\tag{KL}
$$

Plugging in this Equation (KL) into Equation (A5) leads to Equation (A4), as below. This Equation (A4) appears in the main paper (page 9). This labor share is the vital variable for the remaining analysis.

$$
s^L = \frac{1}{1+\left(\frac{1-\Gamma}{\Gamma}\right)\left(\frac{R/A^K}{W/A^L}\right)^{1-\sigma}}
\tag{A4}
$$

The decomposition starts from the percent change in wage bill normalized by polulation (Equation A11-1). Since $\ln\left(\frac{W_t L_t}{N_t}\right)$ can be expressed as $\ln\left(Y_t \sum_i \chi_{it} s^L_{it}\right)$, Equation A11-1 can be decomposed as Equation A11;

$$
\begin{aligned}
&\text{(A11-1)} && \ln\left(\frac{W_t L_t}{N_t}\right) - \ln\left(\frac{W_{t0} L_{t0}}{N_{t0}}\right) \\
&\text{(A11)} && = \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) \\
& && + \ln\left(\sum_i \chi_{it} s_{it}^L\right) - \ln\left(\sum_i \chi_{it0} s_{it0}^L\right) \\
& && = \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) \\
& && + \ln\left(\sum_i \chi_{it} s_{it}^L\right) - \ln\left(\sum_i \chi_{it0} s_{it}^L\right) \\
& && + \ln\left(\sum_i \chi_{it0} s_{it}^L\right) - \ln\left(\sum_i \chi_{it0} s_{it0}^L\right) \\
& && \approx \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) \\
& && + \sum_i \frac{s_{it}^L}{\sum_j \chi_{jt0} s_{jt}^L} (\chi_{it} - \chi_{it0}) \\
& && + \ln\left(\sum_i \chi_{it0} s_{it}^L\right) - \ln\left(\sum_i \chi_{it0} s_{it0}^L\right) \\
&\text{(A11-2)} && \approx \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) \\
& && + \sum_i \frac{s_{it}^L}{\sum_j \chi_{jt0} s_{jt}^L} (\chi_{it} - \chi_{it0}) \\
& && + \sum_i \ell_{it0} (\ln s_{it}^L - \ln s_{it0}^L)
\end{aligned}
$$

The first-order Taylor expansion of the last term of Equation A11-2 yields Equation A11-3; Denote $(1-\sigma)(1-s_{it0}^L)\Big(\ln \frac{W_{it}}{W_{it0}} - \ln \frac{R_{it}}{R_{it0}} - g_{i,t0,t}^A\Big)$ as $\text{Substitution}_{i,t0,t}$, we can rewrite Equation A11-3 as A11-4; Denote $\left(\ln s_{it}^L - \ln s_{it0}^L\right) - \text{Substitution}_{i,t0,t}$ as $\text{ChangeTaskContent}_{i,t0,t}$, we can rewrite Equation A11-4 as A11-4.

$$
\begin{aligned}
\text{(A11-3)} \qquad & \approx \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) \\
& + \sum_i \frac{s^L_{it}}{\sum_j \chi_{jt0} s^L_{jt}} \left(\chi_{it} - \chi_{it0}\right) \\
& + \sum_i \ell_{it0} \Bigg[ (1-\sigma)(1-s^L_{it0}) \Bigg( \ln \frac{W_{it}}{W_{it0}} - \ln \frac{R_{it}}{R_{it0}} - g^A_{i,t0,t} \Bigg) \\
& \qquad + \frac{1-s^L_{it0}}{1-\Gamma_{it0}} (\ln \Gamma_{it} - \ln \Gamma_{it0}) \Bigg] \\
& \approx \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) \\
& + \sum_i \frac{s^L_{it}}{\sum_j \chi_{jt0} s^L_{jt}} \left(\chi_{it} - \chi_{it0}\right) \\
& + \sum_i \ell_{it0} \Bigg[ \text{Substitution}_{i,t0,t} \\
& \qquad + \frac{1-s^L_{it0}}{1-\Gamma_{it0}} (\ln \Gamma_{it} - \ln \Gamma_{it0}) \Bigg] \\
\text{(A11-4)} \qquad & \approx \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) \\
& + \sum_i \frac{s^L_{it}}{\sum_j \chi_{jt0} s^L_{jt}} \left(\chi_{it} - \chi_{it0}\right) \\
& + \sum_i \ell_{it0} \Bigg[ \text{Substitution}_{i,t0,t} \\
& \qquad + \left( \ln s^L_{it} - \ln s^L_{it0} \right) - \text{Substitution}_{i,t0,t} \Bigg] \\
\text{(A11-4)} \qquad & \approx \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) \\
& + \sum_i \frac{s^L_{it}}{\sum_j \chi_{jt0} s^L_{jt}} \left(\chi_{it} - \chi_{it0}\right) \\
& + \sum_i \ell_{it0} \Bigg[ \text{Substitution}_{i,t0,t} \\
& \qquad + \text{ChangeTaskContent}_{i,t0,t} \Bigg] \\
& \approx \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) \\
& + \sum_i \frac{s^L_{it}}{\sum_j \chi_{jt0} s^L_{jt}} \left(\chi_{it} - \chi_{it0}\right) \\
& + \text{Substitution}_{t0,t} \\
& + \sum_i \ell_{it0} \Bigg[ \text{ChangeTaskContent}_{i,t0,t} \Bigg]
\end{aligned}
$$

$\sum_i \ell_{it0}$[ChangeTaskContent$_{i,t0,t}$] can be again decomposed into Equation A14, assuming that over five-year windows, an industry engages in either automation or the creation of new tasks but not in both activities.

$$\text{(A14)} \quad \text{Displacement}_{t-1,t} = \sum_{i \in \mathcal{I}} \ell_{i,t0} \min\left\{0, \frac{1}{5} \sum_{\gamma=t-2}^{t+2} \text{ChangeTaskContent}_{i,\gamma-1,\gamma}\right\}$$

$$\text{Reinstatement}_{t-1,t} = \sum_{i \in \mathcal{I}} \ell_{i,t0} \max\left\{0, \frac{1}{5} \sum_{\gamma=t-2}^{t+2} \text{ChangeTaskContent}_{i,\gamma-1,\gamma}\right\}$$

To sum up, starting from Equation A11-1, it can be decomposed into 1) productivity, 2) composition, 3) subsitution, 4) displacement, and 5) reinstatement effects.

$$\begin{aligned}
\text{(A11-1)} \quad & \ln\left(\frac{W_t L_t}{N_t}\right) - \ln\left(\frac{W_{t0} L_{t0}}{N_{t0}}\right) && \text{[Wage bill per capita]} \\
& \approx \ln\left(\frac{Y_t}{N_t}\right) - \ln\left(\frac{Y_{t0}}{N_{t0}}\right) && \text{[Productivity effect]} \\
& + \sum_i \frac{s_{it}^L}{\sum_j \chi_{jt0} s_{jt}^L}\left(\chi_{it} - \chi_{it0}\right) && \text{[Composition effect]} \\
& + \text{Substitution}_{t0,t} && \text{[Substitution effect]} \\
& + \text{Displacement}_{t0,t} && \text{[Displacement effect (Automation)]} \\
& + \text{Reinstatement}_{t0,t} && \text{[Reinstatement effect (New tasks)]}
\end{aligned}$$